

Gang Qu



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Gang Qu Scholar



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LBC 500, Tulane University, New Orleans, LA 70118

Education

- **Ph.D., Biomedical Engineering, Tulane University** [2018–Now]
GPA:3.81/4.0. Research interests: *Deep learning application on neuroimaging study, medical imaging analysis.*
- **M.Sc., Computer Science, Georgia Institute of Technology** [2021–Now]
GPA:3.5/4.0. Relevant Coursework: *Reinforcement learning, Software development, Database, Computer Networks.*
- **M.Sc., Bioengineering and Biomedical (Minor in Electronic and Computer Engineering) University of Florida** [2016–2018]
GPA:3.83/4.0. Thesis title: *Automatic Pleomorphism Grading For Breast Cancer Image.*
- **B.Ec., Biomedical Engineering, Xi'an Jiaotong University** [2012-2016]
GPA:80/100. Relevant Coursework: *Statistics, Data structure and algorithm, C++ Programming, Signal processing.*

Research Experience

- **AI-Driven Whole-Slide Imaging Digital Health System**
Company: Merck & Co., Inc.
Position: Research Internship
Description: Collaboration with the medical and pathology team at Merck UK.
 1. *Development of an Advanced AI Solution for Automated PD-L1 CPS Analysis in Clinical Applications* [2022]
 - Designed a deep learning-powered AI solution for PD-L1 CPS scoring, inspired by Merck's top-selling and globally recognized KEYTRUDA® (pembrolizumab) anti-PD-1 therapy, which generated close to \$21 billion in 2022.
 - Worked closely with clinical experts to create test scenarios using 302 commercial Stained IHC Whole Slide Images, fine-tuning the AI system's precision with an AUROC of over 0.98, and evaluated its clinical significance by comparing its performance to pathologist evaluations using data from clinical trials.
 - Utilized Python and PyTorch to develop the AI solution, focusing on automated data quality control (such as pen markings, dark areas, tissue folds, etc.), cell segmentation, and predictive modeling with 37,646 annotated cells.

Research Experience (continued)

■ Interpretable Graph deep learning model for multimodal medical imaging analysis

Laboratory: The Multiscale Bioimaging and Bioinformatics Laboratory (MBB), Tulane University, PI: Dr. Yu-Ping Wang.

Position: Research Assistant

Description: Collaboration with TReNDS Center (GSU/Gatech/Emory), DICOIN Lab (Boys Town National Research Hospital), and Mind Research Network for neuroimaging and brain function research, supported by NIH and NSF grants totaling over \$2 million.

1. Graphical Deep Learning for Brain Functional Connectivity Analysis [2018-2020]

- Developed innovative graph-based deep learning methods by applying advanced graph theory for fMRI data analysis and phenotype prediction.
- Employed semi-supervised graph deep learning with Laplacian regularization to address the oversmoothing issue, leveraging the relationships between subjects.

2. Ensemble manifold regularized multi-modal graph convolutional network for cognitive ability prediction [2020-2021]

- Integrated multimodal data to examine associations among various fMRI paradigms and identify key biomarkers.
- Introduced multimodal graph-based deep learning approaches incorporating manifold learning for highly accurate results.

3. Interpretable Cognitive Ability Predictions: A Comprehensive Gated Graph Transformer Framework for Analyzing Functional Brain Networks [2022-2023]

- Implemented prior spatial knowledge and a random-walk diffusion strategy to simultaneously capture complex structural and functional relationships between brain regions.
- Applied attention mechanisms for learning multi-view node feature embeddings and dynamically assigning propagation weights, allowing for the identification of significant functional brain network biomarkers and enhancing result interpretability.

■ Rule-based End-to-End Lung Cancer Classification on Whole Slide Images (master thesis), University of Florida

Position: Research Assistant

Description: Collaboration with pathologists and medical doctors for clinical applications.

1. Graph temporal ensembling based semi-supervised convolutional neural network with noisy labels for histopathology image analysis [2017-2018]

- Designed a rule-based CNN model (Nottingham Histologic Grade) for classifying SCC and ADC breast cancer cells, leveraging TensorFlow and PyTorch frameworks.
- Employed advanced pre-processing techniques for whole-slide images, including image augmentation and data normalization, to minimize potential biases in the dataset.

Work Experience

2022 Summer ■ Research Internship, Merck & Co., Inc.

2018-Now ■ Research Assistant, Tulane University

Work Experience (continued)

- 2021–Now  **Lab technician, MBB Lab & CBG Center, Tulane University**
- 2017–2018  **Research Assistant, University of Florida**

Research Publications

Journal Articles

- 1 **G. Qu**, W. Hu, L. Xiao, *et al.*, “Brain functional connectivity analysis via graphical deep learning,” *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 5, pp. 1696–1706, 2022.
- 2 **G. Qu**, L. Xiao, W. Hu, *et al.*, “Ensemble manifold regularized multi-modal graph convolutional network for cognitive ability prediction,” *IEEE Transactions on Biomedical Engineering*, vol. 68, no. 12, pp. 3564–3573, 2021.
- 3 W. Yan, **G. Qu**, W. Hu, *et al.*, “Deep learning in neuroimaging: Promises and challenges,” *IEEE Signal Processing Magazine*, vol. 39, no. 2, pp. 87–98, 2022.
- 4 X. Shi, H. Su, F. Xing, Y. Liang, **G. Qu**, and L. Yang, “Graph temporal ensembling based semi-supervised convolutional neural network with noisy labels for histopathology image analysis,” *Medical image analysis*, vol. 60, p. 101624, 2020.
- 5 L. Xiao, B. Cai, **G. Qu**, *et al.*, “Distance correlation-based brain functional connectivity estimation and non-convex multi-task learning for developmental fmri studies,” *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 10, pp. 3039–3050, 2022.
- 6 W. Hu, X. Meng, Y. Bai, A. Zhang, **G. Qu**, *et al.*, “Interpretable multimodal fusion networks reveal mechanisms of brain cognition,” *IEEE Transactions on Medical Imaging*, vol. 40, no. 5, pp. 1474–1483, 2021.
- 7 J. Wang, L. Xiao, W. Hu, **G. Qu**, *et al.*, “Functional network estimation using multigraph learning with application to brain maturation study,” *Human Brain Mapping*, vol. 42, no. 9, pp. 2880–2892, 2021.
- 8 A. Orlichenko, **G. Qu**, G. Zhang, *et al.*, “Latent similarity identifies important functional connections for phenotype prediction,” *IEEE Transactions on Biomedical Engineering*, 2023, early access.

Conference Proceedings

- 1 **G. Qu**, W. Hu, L. Xiao, and Y.-P. Wang, “A graph deep learning model for the classification of groups with different IQ using resting state fMRI,” A. Krol and B. S. Gimi, Eds., International Society for Optics and Photonics, vol. 11317, SPIE, 2020, 113170A.
- 2 A. Orlichenko, **G. Qu**, and Y.-P. Wang, “Phenotype guided interpretable graph convolutional network analysis of fMRI data reveals changing brain connectivity during adolescence,” B. S. Gimi and A. Krol, Eds., International Society for Optics and Photonics, vol. 12036, SPIE, 2022, p. 1203612.

Presentations

Conference Presentation

-  **OHBM Annual Meeting**, Montréal, Québec, Canada. [2023]
Exploring General Intelligence via Gated Graph Transformer in fMRI Functional Connectivity Studies
-  **SPIE Medical Imaging**, Houston, TX, USA. [2020]
A graph deep learning model for the classification of groups with different IQ using resting state fMRI

Presentations (continued)

Campus Presentation

- **Virtual dev-CoG/dev-MIND Meeting**, Virtual Presentation. [2021]
Structure-enriched Collaborative Regression (SCoRe) and its Application to Brain-related Study
- **Virtual dev-CoG/dev-MIND Meeting**, Virtual Presentation. [2020]
Ensemble manifold based regularized multi-modal graph convolutional network for cognitive ability prediction

Service To Profession

Conference Presentation

- Reviewer for Medical Image Analysis (MedIA)
- Reviewer for Computers in Biology and Medicine

Skills

- **Programming:** Python, TensorFlow, Pytorch, Java, MATLAB, SQL, C/C++, R
- **Database:** MySQL, EER Diagram Design
- **Language:** Chinese (native), English (fluent)
- **Others:** Git, Microsoft Office, $\text{\LaTeX}$

Membership

- IEEE Student Membershipt [2022-Now]
- Member of SPIE, the international society for optics and photonics [2020-2021]

Honors & Certifications

Awards and Achievements

- **Academic Honor Achievement Award**, University of Florida [2017]
- **Academic Honor Achievement Award**, University of Florida [2016]

Certification

- **Machine Learning**. Awarded by Coursera [2020]
- **Natural Language Processing Specification**. Awarded by Coursera [2020]
- **Deep Learning Specification**. Awarded by Coursera [2020]